1 **(a)**	force per unit mass	B1	[1]
(b) **(i)**	radius/diameter/size (of Proxima Centauri) $\ll$ /is <u>much</u> less than 4.0×10^{13} km/separation (of Sun and star) *or* (because) it is a <u>uniform</u> sphere	B1	[1]
(ii) **1.**	field strength $= GM/x^2$		
	$= (6.67 \times 10^{-11} \times 2.5 \times 10^{29}) / (4.0 \times 10^{13} \times 10^3)^2$	C1	
	$= 1.0 \times 10^{-14}\ \mathrm{N\,kg^{-1}}$	A1	[2]
2.	force $=$ field strength $\times$ mass		
	$= 1.0 \times 10^{-14} \times 2.0 \times 10^{30}$	C1	
	or		
	force $= GMm/x^2$		
	$= (6.67 \times 10^{-11} \times 2.5 \times 10^{29} \times 2.0 \times 10^{30}) / (4.0 \times 10^{13} \times 10^3)^2$	(C1)	
	$= 2.0 \times 10^{16}\ \mathrm{N}$	A1	[2]
(c)	force (of 2×10^{16} N) would have little effect on (large) <u>mass</u> of Sun	B1	
	would cause an acceleration of Sun of $1.0 \times 10^{-14}\ \mathrm{m\,s^{-2}}$/very small/negligible acceleration	B1	[2]
	or		
	many stars all around the Sun	(B1)	
	net effect of forces/fields is zero	(B1)	